

# Recommendation Model Training & Evaluation Report

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## 1. Dataset Summary

| Metric                      | Value                                 |
|-----------------------------|---------------------------------------|
| Dataset Loaded Successfully | Yes (recommendation_dataset.csv)      |
| Dataset Shape               | 100,000 rows × 21 columns             |
| Train / Test Split          | 80% (80,000) / 20% (20,000)           |
| Target Column               | recommendation (8 Multi-class Target) |
| Feature Scaling             | StandardScaler fitted on X_train      |

## 2. Model Comparison Table

| Model             | Accuracy | Precision | Recall | Specificity | F1     | ROC AUC | AP     | MCC    |
|-------------------|----------|-----------|--------|-------------|--------|---------|--------|--------|
| Decision Tree     | 0.4339   | 0.4330    | 0.4339 | 0.9189      | 0.4325 | 0.7985  | 0.3836 | 0.3518 |
| Random Forest     | 0.5058   | 0.5061    | 0.5058 | 0.9292      | 0.5043 | 0.8824  | 0.5185 | 0.4344 |
| Gradient Boosting | 0.5228   | 0.5222    | 0.5228 | 0.9317      | 0.5220 | 0.8973  | 0.5548 | 0.4539 |

## 3. Best Model & Saved Files

| Attribute            | Value                                     |
|----------------------|-------------------------------------------|
| Best Selected Model  | Gradient Boosting                         |
| Best F1 Score        | 0.5220                                    |
| Best Accuracy        | 0.5228                                    |
| Best MCC             | 0.4539                                    |
| Saved Model          | models/recommendation/best_model.pkl      |
| Saved Scaler         | models/recommendation/scaler.pkl          |
| Saved Metadata       | models/recommendation/metadata.json       |
| Saved Model Metadata | models/recommendation/model_metadata.json |